

Operations & Maintenance Runbook

ALRP Production Environment

Prepared by: Platform Operations

Baseline: NIST SP 800-53 Rev. 5 -- Moderate

Classification: Controlled Unclassified Information (CUI)

Status: Draft for pre-adjudication review

1. Operational Overview

This runbook documents the day-to-day operational procedures that implement and sustain the security controls of the ALRP production environment. It is maintained by Platform Operations and is the authoritative reference for routine maintenance, monitoring, and recovery activities.

2. Environment

Production runs as containerized services behind a managed load balancer, with data persisted to an encrypted managed database and an object store. Infrastructure is defined as code and deployed through the controlled pipeline described in the Configuration Management Plan.

3. Control Implementations

AC-17 Remote Access

Remote access extends the authorization boundary to authenticated users connecting from outside the corporate network.

Remote-access entitlements and VPN allow-lists are reviewed quarterly by the network security team.

All remote sessions traverse the corporate VPN with enforced multi-factor authentication and are logged centrally.

AC-2 Account Management

Account management governs the full lifecycle of identities that can access the system, from provisioning through disablement.

Privileged, standard, and service accounts are reviewed weekly by the ISSO to confirm continued need and least-privilege alignment.

Provisioning is initiated from an approved request and enforced through the enterprise identity provider; the ISSO is accountable for the control.

AU-11 Audit Record Retention

Audit retention preserves the evidentiary record required for after-the-fact investigation and accountability.

Audit-retention configuration and storage integrity are validated quarterly against the documented policy.

Records are written to an immutable, access-controlled store with integrity protection enabled.

AU-6 Audit Record Review

Audit review converts raw event data into actionable detections by surfacing anomalous and unauthorized behavior.

Audit logs are reviewed for anomalous and unauthorized activity daily by the on-duty SOC analyst.

Events are aggregated into the SIEM and correlated against detection content aligned to the MITRE ATT&CK matrix.

CM-3 Configuration Change Control

Configuration change control prevents unauthorized or unreviewed modifications from degrading the security baseline.

Proposed configuration changes are reviewed and dispositioned by the Change Advisory Board monthly.

Each change carries a documented justification, rollback plan, and security impact analysis before approval.

CP-9 System Backup

System backup protects the availability of data and configuration against corruption, ransomware, and hardware loss.

Backup restoration is tested monthly into an isolated recovery environment to confirm recoverability.

Backups are encrypted, replicated across regions, and protected against deletion before their retention expires.

RA-5 Vulnerability Monitoring and Scanning

Vulnerability scanning identifies exploitable weaknesses before an adversary can leverage them.

Authenticated vulnerability scans are executed monthly across every asset inside the authorization boundary.

Scan results are risk-ranked and remediated within timelines keyed to severity, with exceptions formally accepted.

SI-4 System Monitoring

System monitoring provides the detective controls that surface intrusions and policy violations in near-real time.

Security-monitoring detections and dashboards are reviewed weekly by the Security Operations Center.

Detections forward to an alerting pipeline that pages the on-call analyst, who triages within the documented window.

4. Supporting Controls

MA-2 Controlled Maintenance

Maintenance activities are scheduled, approved, and logged, and maintenance personnel are supervised in accordance with their authorization.

SC-7 Boundary Protection

Traffic entering the boundary passes through a web application firewall and a load balancer that terminates transport encryption; egress is restricted to documented service endpoints.